

PERCENTAGES

Full Solutions & Explanations

DOCUMENT 2 — SOLUTIONS & EXPLANATIONS

Total Questions: 40

Q1. Correct Answer: **C • EASY**

STEP-BY-STEP SOLUTION:

To find a percent of a number, convert the percent to a decimal and multiply.

$$35\% = 0.35.$$

$$0.35 \times 220 = 77.$$

WHY EACH OPTION IS WRONG:

- A)** This results from a decimal-placement error, off by a factor of 10 from the correct answer.
- B)** This computes $220 - 77$ (the REMAINING amount after removing 35%) instead of the 35% portion itself.
- C)** Correct. $0.35 \times 220 = 77$.
- D)** This adds 35 directly to 220 ($220 + 35 = 255$), treating the percent as if it were a flat number instead of a rate to multiply by.

TRAP ALERT

Treating the percent value as a number to add or subtract directly, instead of converting it to a decimal and multiplying.

KEY INSIGHT

"X% of a number" always means $(X/100) \times \text{number}$ — convert to a decimal first, then multiply.

FAST METHOD

Move the decimal point two places left to convert the percent ($35\% \rightarrow 0.35$), then multiply directly.

Q2. Correct Answer: **D • EASY**

STEP-BY-STEP SOLUTION:

First find the amount of the increase: $25\% \text{ of } \$80 = 0.25 \times 80 = \20 .

Add this increase to the original price: $\$80 + \$20 = \$100$.

WHY EACH OPTION IS WRONG:

- A)** This reports only the AMOUNT of the increase (\$20), not the new total price the question asks for.
- B)** This results from an arithmetic slip while adding the increase to the original price.
- C)** This results from using an incorrect percent (such as 5%) instead of the given 25%.
- D)** Correct. $\$80 + (0.25 \times 80) = \$80 + \$20 = \100 .

TRAP ALERT

Stopping after finding the increase amount and forgetting that the question asks for the NEW price, which requires adding that amount back to the original.

KEY INSIGHT

Percent increase problems require two steps: find the change amount (percent $\times$ original), then add it to the original value to get the new total.

FAST METHOD ⚡

For a quick shortcut, multiply the original value directly by (1 + percent as a decimal): $80 \times 1.25 = 100$.

Q3. Correct Answer: **A • EASY**

STEP-BY-STEP SOLUTION:

First find the amount of the decrease: 20% of \$150 = $0.20 \times 150 = \$30$.

Subtract this decrease from the original price: $\$150 - \$30 = \$120$.

WHY EACH OPTION IS WRONG:

- A)** Correct. $\$150 - (0.20 \times 150) = \$150 - \$30 = \120 .
- B)** This reports only the AMOUNT of the decrease (\$30), not the new price the question asks for.
- C)** This results from an arithmetic slip while subtracting.
- D)** This results from using an incorrect percent (such as 16.7%) instead of the given 20%.

TRAP ALERT 🚧

Stopping after finding the decrease amount and forgetting that the question asks for the NEW (reduced) price.

KEY INSIGHT 💡

Percent decrease problems require finding the change amount (percent $\times$ original), then subtracting it from the original value.

FAST METHOD ⚡

Multiply the original value directly by (1 – percent as a decimal): $150 \times 0.80 = 120$.

Q4. Correct Answer: **B • EASY**

STEP-BY-STEP SOLUTION:

To find what percent one number is of another, divide the part by the whole and convert to a percent: $(45/180) \times 100\%$.

$$45/180 = 0.25.$$

$$0.25 \times 100\% = 25\%.$$

WHY EACH OPTION IS WRONG:

- A)** This results from a decimal-placement error, off by a factor of 10 from the correct answer.
- B)** Correct. $45/180 = 0.25 = 25\%$.
- C)** This results from an arithmetic slip while dividing.
- D)** This inverts the fraction, computing $180/45$ instead of $45/180$, giving the reciprocal relationship instead.

TRAP ALERT 🚧

Dividing the numbers in the wrong order (the "whole" divided by the "part" instead of the "part" divided by the "whole"), which inverts the entire answer.

KEY INSIGHT 💡

"A is what percent of B" always translates to $(A/B) \times 100\%$ — the number right after "of" goes in the denominator.

FAST METHOD ⚡

Simplify the fraction first if possible: $45/180$ reduces to $1/4$, which is immediately recognizable as 25%.

Q5. Correct Answer: **26** • EASY 🖊️ GRID-IN

STEP-BY-STEP SOLUTION:

Convert the percent to a decimal: $40\% = 0.40$.

Multiply: $0.40 \times 65 = 26$.

COMMON WRONG ANSWERS & WHY:

- Gridding 2.6 — a decimal-placement error, off by a factor of 10.
- Gridding 25 — an arithmetic slip while multiplying.

TRAP ALERT 🚨

Misplacing the decimal point while converting the percent or while multiplying, especially with two-digit percents and two-digit numbers together.

KEY INSIGHT 💡

Converting a percent to a decimal always means dividing by 100 (moving the decimal point two places left): 40% becomes 0.40, not 4.0 or 0.04.

FAST METHOD ⚡

Recognize that 40% is the same as $2/5$, so $65 \times 2/5 = 130/5 = 26$ — sometimes a fraction shortcut is faster than decimal multiplication.

Q6. Correct Answer: **C** • EASY

STEP-BY-STEP SOLUTION:

Convert the percent to a decimal: $12\% = 0.12$.

Multiply: $0.12 \times 250 = 30$.

WHY EACH OPTION IS WRONG:

- A)** This results from a decimal-placement error, off by a factor of 10.
- B)** This results from using an incorrect percent (such as 10%) instead of the given 12%.
- C)** Correct. $0.12 \times 250 = 30$.
- D)** This adds 12 directly to 250 ($250 + 12 = 262$), treating the percent as a flat number instead of a rate to multiply by.

TRAP ALERT 🚨

Adding the percent value directly to the number instead of converting it to a decimal and multiplying.

KEY INSIGHT 💡

A percent is always a rate to be multiplied, never a quantity to be added or subtracted directly (unless it's being applied as part of a separate increase/decrease step).

FAST METHOD ⚡

Break 12% into 10% + 2%: 10% of 250 is 25, and 2% of 250 is 5, so $25 + 5 = 30$.

Q7. Correct Answer: **D • EASY**

STEP-BY-STEP SOLUTION:

Let the unknown number be n . The given information translates to: $0.20 \times n = 35$.
Divide both sides by 0.20: $n = 35/0.20$.
 $n = 175$.

WHY EACH OPTION IS WRONG:

- A)** This multiplies instead of dividing ($35 \times 0.20 = 7$), going in the wrong direction.
- B)** This adds the two given numbers ($35 + 20 = 55$) instead of setting up and solving the percent equation.
- C)** This subtracts the two given numbers ($35 - 20 = 15$) instead of setting up and solving the percent equation.
- D)** Correct. $0.20n = 35$ leads to $n = 35/0.20 = 175$.

TRAP ALERT 🚧

Multiplying by the percent instead of dividing by it — when you're given a PART and a percent and asked for the WHOLE, division (not multiplication) is required.

KEY INSIGHT 💡

"X% of a number is Y" translates to the equation $(X/100) \times n = Y$; solving for n always requires dividing Y by the decimal form of $X\%$.

FAST METHOD ⚡

Since 20% is $1/5$, and $1/5$ of the number is 35, the whole number is $5 \times 35 = 175$.

Q8. Correct Answer: **A • EASY**

STEP-BY-STEP SOLUTION:

First find the discount amount: $15\% \text{ of } \$60 = 0.15 \times 60 = \9 .
Subtract this discount from the original price: $\$60 - \$9 = \$51$.

WHY EACH OPTION IS WRONG:

- A)** Correct. $\$60 - (0.15 \times 60) = \$60 - \$9 = \51 .
- B)** This reports only the discount AMOUNT (\$9), not the actual sale price.
- C)** This results from using an incorrect percent (such as 25%) instead of the given 15%.
- D)** This adds the discount instead of subtracting it, increasing the price rather than reducing it.

TRAP ALERT 🚧

Adding the discount amount instead of subtracting it, or stopping after computing only the discount amount instead of the final sale price.

KEY INSIGHT 💡

"Off" always signals subtraction: find the discount amount first (percent $\times$ original price), then subtract it from the original price.

FAST METHOD ⚡

Multiply directly by $(1 - \text{discount as a decimal})$: $60 \times 0.85 = 51$.

Q9. Correct Answer: **54** • EASY  GRID-IN

STEP-BY-STEP SOLUTION:

Find the increase amount: 35% of 40 = $0.35 \times 40 = 14$.

Add this to the original value: $40 + 14 = 54$.

COMMON WRONG ANSWERS & WHY:

- Gridding 14 — reporting only the increase AMOUNT instead of the new total value.
- Gridding 75 — an arithmetic or setup error while combining the original value and the increase.

TRAP ALERT

Stopping after computing the increase amount and forgetting to add it back to the original value to find the actual new total.

KEY INSIGHT

Percent increase always means: new value = original value + (percent $\times$ original value), or equivalently, original $\times$ (1 + percent as a decimal).

FAST METHOD

Multiply directly: $40 \times 1.35 = 54$, combining both steps into one calculation.

Q10. Correct Answer: **B** • EASY

STEP-BY-STEP SOLUTION:

Find the tax amount: 8% of \$40 = $0.08 \times 40 = \$3.20$.

Add this tax to the original price: $\$40 + \$3.20 = \$43.20$.

WHY EACH OPTION IS WRONG:

- A)** This reports only the tax AMOUNT (\$3.20), not the total cost including the original price.
- B)** Correct. $\$40 + (0.08 \times 40) = \$40 + \$3.20 = \43.20 .
- C)** This results from a decimal-placement error while computing the tax (treating 8% as 0.08% instead of 0.08).
- D)** This results from using an incorrect percent (such as 20%) instead of the given 8%.

TRAP ALERT

Reporting only the tax amount instead of the TOTAL cost, which requires adding the tax back to the original price.

KEY INSIGHT

"Total cost including tax" always means original price + tax amount, where tax amount = tax rate $\times$ original price.

FAST METHOD

Multiply directly by (1 + tax rate as a decimal): $40 \times 1.08 = 43.20$.

Q11. Correct Answer: **C** • MEDIUM

STEP-BY-STEP SOLUTION:

Let the original price be \$100 for simplicity. After a 20% increase: $100 \times 1.20 = \$120$.

After a 20% decrease applied to this NEW price: $120 \times 0.80 = \$96$.

The overall change is from \$100 to \$96, a decrease of \$4, which is a 4% decrease from the original — written as -4%.

WHY EACH OPTION IS WRONG:

- A)** This assumes the two equal-percent changes cancel out perfectly, but they don't — the 20% decrease is applied to a LARGER number (120) than the 20% increase was applied to (100), so the dollar amounts aren't equal.
- B)** This gets the correct magnitude (4%) but the wrong sign, mistakenly treating the net result as an increase.
- C)** Correct. $100 \times 1.20 \times 0.80 = 96$, which is a 4% decrease (-4%) from the original 100.
- D)** This simply adds the two percents together as if they were both decreases (-20% + -20% = -40%), ignoring that one was actually an increase and that percent changes don't add linearly anyway.

TRAP ALERT

Assuming that an X% increase followed by an X% decrease (or vice versa) always returns to the original value — it never does (except when X=0), because the second percentage is always applied to a DIFFERENT base value than the first.

KEY INSIGHT

Successive percent changes are never simply additive. A percent increase followed by an equal percent decrease always results in a net DECREASE, because the decrease is calculated from a larger base than the increase was.

FAST METHOD

Multiply the two factors directly: $(1.20)(0.80) = 0.96$, which is 4% below the original 1.00 — the net change is always $(1+a)(1-a)-1 = -a^2$ for equal-and-opposite percents, always negative.

Q12. Correct Answer: **D** • MEDIUM

STEP-BY-STEP SOLUTION:

Let the original price be p . After a 15% increase, the new price is $p \times 1.15$.
Set this equal to the given final price: $1.15p = 138$.
Divide both sides by 1.15: $p = 138/1.15 = 120$.

WHY EACH OPTION IS WRONG:

- A)** This incorrectly subtracts 15% from the final price (138×0.85) instead of dividing by 1.15, which is not the correct way to reverse a percent increase.
- B)** This multiplies by 1.15 again instead of dividing by it, moving in the wrong direction entirely.
- C)** This results from an arithmetic slip while dividing.
- D)** Correct. $138/1.15 = 120$.

TRAP ALERT

Trying to "undo" a 15% increase by simply taking 15% off the FINAL price, instead of recognizing that reversing a percent increase requires DIVIDING by $(1 + \text{the percent as a decimal})$, not subtracting a percent of the final value.

KEY INSIGHT

To reverse a percent increase and find the original value, divide the final value by $(1 + \text{percent as a decimal})$ — never just subtract that same percent from the final value, since percents of different base values aren't interchangeable.

FAST METHOD

Set up the equation $(\text{original}) \times (1 + \text{percent}) = \text{final}$, then solve by dividing: $138 \div 1.15 = 120$.

Q13. Correct Answer: **28** • MEDIUM  GRID-IN

STEP-BY-STEP SOLUTION:

Percent increase is calculated as $(\text{change in value})/(\text{original value}) \times 100\%$, where the ORIGINAL value here is March's sales.

The change in value is $320 - 250 = 70$ books.

Percent more = $70/250 \times 100\% = 0.28 \times 100\% = 28\%$.

COMMON WRONG ANSWERS & WHY:

- Gridding 21.9 — using April's sales (320) as the denominator instead of March's sales (250), which represents the wrong base for this comparison.
- Gridding 70 — reporting the raw difference in books sold instead of converting that difference to a percent of the original value.

TRAP ALERT

Using the WRONG base value in the denominator — "April's sales were what percent more than March's" specifically requires March's value (the one being compared FROM) as the base, not April's.

KEY INSIGHT

In "A is what percent more than B" problems, B (the value coming after "than") is always the base for the percent calculation: $\text{percent} = (A-B)/B \times 100\%$.

FAST METHOD

Identify the base value first (the one right after "than"), then compute $(\text{difference})/(\text{base}) \times 100\%$ directly: $70/250 = 0.28 = 28\%$.

Q14. Correct Answer: **A** • MEDIUM

STEP-BY-STEP SOLUTION:

First apply the 25% discount: $\$200 \times 0.75 = \150 (the discounted price).

Then apply the 8% tax to this DISCOUNTED price, not the original: $\$150 \times 1.08 = \162 .

WHY EACH OPTION IS WRONG:

- A)** Correct. $\$200 \times 0.75 \times 1.08 = \$150 \times 1.08 = \$162$.
- B)** This incorrectly applies the 8% tax to the ORIGINAL \$200 price (adding \$16) instead of to the discounted \$150 price, then adds that tax to the discounted price.
- C)** This correctly finds the discounted price but forgets to add the tax entirely.
- D)** This treats the discount and tax as a single combined 33% reduction ($25\%+8\%$) applied once, instead of two separate sequential multiplications.

TRAP ALERT

Applying the sales tax to the ORIGINAL price instead of the already-discounted price, or trying to combine the discount and tax percentages by simple addition instead of applying them as two separate sequential steps.

KEY INSIGHT

When a discount and a tax are both applied, they're applied SEQUENTIALLY, each to the result of the previous step — the tax is calculated on the discounted price, not the original price.

FAST METHOD ⚡

Multiply all the factors together in one chain: $200 \times 0.75 \times 1.08 = 162$ — the order of multiplication doesn't actually matter here, but each step must use the correct decimal factor.

Q15. Correct Answer: **B** • MEDIUM**STEP-BY-STEP SOLUTION:**

Percent error is calculated as $|\text{measured value} - \text{actual value}| / (\text{actual value}) \times 100\%$.

$$|48 - 50| = 2.$$

$$\text{Percent error} = 2/50 \times 100\% = 0.04 \times 100\% = 4\%.$$

WHY EACH OPTION IS WRONG:

- A)** This results from a decimal-placement error, off by a factor of roughly 4 from the correct answer.
- B)** Correct. $|48-50|/50 \times 100\% = 2/50 \times 100\% = 4\%$.
- C)** This results from an unnecessary additional division by 2, halving the correct percent error.
- D)** This computes the ratio of measured to actual ($48/50 = 96\%$) directly, confusing that ratio with the percent error itself instead of finding the DIFFERENCE first.

TRAP ALERT 🚧

Computing the ratio of the measured value to the actual value directly ($48/50$) and mistaking that for the percent error, instead of first finding the DIFFERENCE between the two values and dividing that difference by the actual value.

KEY INSIGHT 💡

Percent error always starts with a SUBTRACTION (the absolute difference between measured and actual), divided by the actual (true) value — it is never just the ratio of the two raw values to each other.

FAST METHOD ⚡

Find the absolute difference first ($|48-50|=2$), then divide by the ACTUAL value ($2/50=0.04=4\%$) — always use the actual/true value as the denominator, not the measured one.

Q16. Correct Answer: **C** • MEDIUM**STEP-BY-STEP SOLUTION:**

Work through this step by step, applying one percent at a time.

First find 40% of 500: $0.40 \times 500 = 200$.

Then find 30% of that result: $0.30 \times 200 = 60$.

WHY EACH OPTION IS WRONG:

- A)** This incorrectly ADDS the two percents together ($30\%+40\%=70\%$) and applies that combined percent to 500 in one step, instead of applying the two percents sequentially.
- B)** This correctly computes the first step ($40\% \text{ of } 500 = 200$) but stops there, forgetting to apply the second percent (30%) at all.
- C)** Correct. $0.40 \times 500 = 200$, then $0.30 \times 200 = 60$.
- D)** This results from a decimal-placement error in one of the two multiplication steps.

TRAP ALERT 🚧

Adding the two percents together before applying them to the original number, instead of applying them one after another in sequence — "X% of Y% of a number" is never the same as "(X+Y)% of that number."

KEY INSIGHT 💡

"X% of Y% of a number" requires two SEPARATE sequential multiplications, applied one at a time, each to the result of the previous step — never combine the percents by addition first.

FAST METHOD ⚡

Multiply the two decimal forms together first, then apply to the original number in one final step:
 $0.30 \times 0.40 = 0.12$, then $0.12 \times 500 = 60$.

Q17. Correct Answer: **250** • MEDIUM ✎ GRID-IN

STEP-BY-STEP SOLUTION:

Let the original price be p . After a 12% decrease, the new price is $p \times 0.88$ (since $100\% - 12\% = 88\%$).

Set this equal to the given final price: $0.88p = 220$.

Divide both sides by 0.88: $p = 220/0.88 = 250$.

COMMON WRONG ANSWERS & WHY:

- Gridding 246.40 — incorrectly adding 12% to the final price (220×1.12) instead of dividing by 0.88, which is not the correct way to reverse a percent decrease.
- Gridding 196.90 (approximately) — multiplying by 0.88 again instead of dividing by it, moving in the wrong direction.

TRAP ALERT 🚧

Trying to reverse a percent decrease by simply ADDING that same percent back to the final value, instead of dividing by $(1 - \text{the percent as a decimal})$.

KEY INSIGHT 💡

To reverse a percent decrease and find the original value, divide the final value by $(1 - \text{percent as a decimal})$ — adding the percent back to the final value does NOT correctly undo the original decrease.

FAST METHOD ⚡

Set up $(\text{original}) \times (1 - \text{percent}) = \text{final}$, then divide to solve: $220 \div 0.88 = 250$.

Q18. Correct Answer: **D** • MEDIUM

STEP-BY-STEP SOLUTION:

Since every person prefers either tea or coffee, the percent who prefer coffee is the complement of the percent who prefer tea: $100\% - 45\% = 55\%$.

Find 55% of the total 300 people: $0.55 \times 300 = 165$.

WHY EACH OPTION IS WRONG:

- A)** This applies the TEA percentage (45%) directly to find the coffee count, instead of first finding the complementary percentage (55%).
- B)** This simply reports the given percent number (45) instead of computing an actual count of people.

C) This subtracts 45 (treating it as a raw count rather than a percent) from 300, instead of properly using percentages throughout.

D) Correct. $100\% - 45\% = 55\%$ prefer coffee; $0.55 \times 300 = 165$.

TRAP ALERT 🚫

Applying the given percentage directly to the wrong group, forgetting that when only two categories exist, the second category's percentage is the **COMPLEMENT** (100% minus the given percent), not the same percentage.

KEY INSIGHT 💡

When a survey or scenario splits a population into exactly two complementary categories, finding one category's count from the other always requires first computing 100% minus the given percentage.

FAST METHOD ⚡

Subtract from 100% first ($100 - 45 = 55$), then multiply by the total: $0.55 \times 300 = 165$.

Q19. Correct Answer: **A** • MEDIUM

STEP-BY-STEP SOLUTION:

Markup percent is calculated as $(\text{selling price} - \text{cost}) / (\text{cost}) \times 100\%$, since the question specifies the markup should be based on cost.

Selling price – cost = $58 - 40 = 18$.

Markup percent = $18/40 \times 100\% = 0.45 \times 100\% = 45\%$.

WHY EACH OPTION IS WRONG:

A) Correct. $(58 - 40) / 40 \times 100\% = 18/40 \times 100\% = 45\%$.

B) This uses the **SELLING PRICE** (58) as the base instead of the cost (40), which the question specifically says **NOT** to do.

C) This reports only the raw dollar difference (\$18), not the percent the question actually asks for.

D) This reports the ratio of selling price to cost ($58/40 = 145\%$) without subtracting 100% first to isolate just the markup portion.

TRAP ALERT 🚫

Using the selling price as the base for the percent calculation instead of the cost, even when the question explicitly specifies which base to use — or forgetting to subtract before dividing, which conflates the **TOTAL** price ratio with the markup percent itself.

KEY INSIGHT 💡

Markup percent based on cost is always $(\text{selling price} - \text{cost}) / \text{cost} \times 100\%$ — always read carefully which value the question specifies as the base, since markup-on-cost and markup-on-selling-price give different percentages for the same transaction.

FAST METHOD ⚡

Find the dollar markup first ($58 - 40 = 18$), then divide by the **SPECIFIED** base — here, cost (40): $18/40 = 0.45 = 45\%$.

Q20. Correct Answer: **B** • MEDIUM

STEP-BY-STEP SOLUTION:

Let the original salary be \$100 for simplicity. After the first 10% increase: $100 \times 1.10 = \$110$.

After the second 10% increase, applied to this NEW salary: $110 \times 1.10 = \$121$.
The total change is from \$100 to \$121, which is a 21% increase overall.

WHY EACH OPTION IS WRONG:

- A)** This simply adds the two 10% increases together ($10\%+10\%=20\%$), ignoring that the second increase is applied to a LARGER base than the first.
- B)** Correct. $1.10 \times 1.10 = 1.21$, a 21% total increase.
- C)** This accounts for only one of the two increases, forgetting the second 10% increase entirely.
- D)** This results from a different arithmetic slip while combining the two increases.

TRAP ALERT

Adding two successive percent increases directly ($10\%+10\%=20\%$) instead of recognizing that the second increase compounds on the already-increased value, making the true total slightly MORE than the simple sum.

KEY INSIGHT

Two successive percent increases of the same size always compound to slightly MORE than double that percent (here, 21% rather than 20%), because the second increase is calculated from a larger starting point than the first.

FAST METHOD

Multiply the two growth factors directly: $1.10 \times 1.10 = 1.21$, which is a 21% total increase — never just add the two percents.

Q21. Correct Answer: **C** • MEDIUM

STEP-BY-STEP SOLUTION:

Percent less is calculated as $(\text{difference})/(\text{original value}) \times 100\%$, where the ORIGINAL value here is City A's population (the one being compared FROM).

The difference is $50,000 - 35,000 = 15,000$.

Percent less = $15,000/50,000 \times 100\% = 0.30 \times 100\% = 30\%$.

WHY EACH OPTION IS WRONG:

- A)** This uses the SUM of the two populations (85,000) as the denominator instead of City A's population alone.
- B)** This uses City B's population (35,000) as the denominator instead of City A's, which represents the wrong base for this comparison.
- C)** Correct. $(50,000-35,000)/50,000 \times 100\% = 15,000/50,000 \times 100\% = 30\%$.
- D)** This misreads the raw difference (15,000) as if it directly translated to "15%" without actually dividing by the base population at all.

TRAP ALERT

Using the wrong base value in the denominator (City B's population, or the sum of both) instead of City A's population, which is the correct reference point since the question asks how much LESS City B is than City A.

KEY INSIGHT

In "A is what percent less than B" problems, B is always the base for the calculation: $\text{percent} = (B-A)/B \times 100\%$ — always identify which value comes right after "than" to find the correct base.

FAST METHOD

Identify the base (the value right after "than"), then compute $(\text{difference})/(\text{base}) \times 100\%$:
 $15,000/50,000 = 0.30 = 30\%$.

Q22. Correct Answer: **57.6** • MEDIUM  GRID-IN

STEP-BY-STEP SOLUTION:

Apply the first discount: $\$80 \times 0.90 = \72 (the price after the first discount).

Apply the second discount to this NEW price: $\$72 \times 0.80 = \57.60 .

COMMON WRONG ANSWERS & WHY:

- Gridding 56 — incorrectly combining the two discounts by simple addition ($10\%+20\%=30\%$ off), then applying that combined 30% to the original \$80 directly: $80 \times 0.70 = 56$.
- Gridding 51.20 — applying the discounts in a way that miscalculates one of the two sequential steps.

TRAP ALERT

Adding the two discount percentages together ($10\%+20\%=30\%$) and applying that single combined discount to the original price, instead of applying each discount sequentially to the progressively reduced price — these give different results, and the sequential method is always the correct one for "successive" discounts.

KEY INSIGHT

Successive (stacked) discounts must be applied one after another, each to the result of the previous discount — never combined by simple addition, since the second discount is calculated on a SMALLER base than the original price.

FAST METHOD

Multiply all the remaining-value factors together in one chain: $80 \times 0.90 \times 0.80 = 57.60$.

Q23. Correct Answer: **D** • MEDIUM

STEP-BY-STEP SOLUTION:

Percent decrease is calculated as $(\text{original} - \text{new})/(\text{original}) \times 100\%$.

The decrease amount is $80 - 68 = 12$.

Percent decrease = $12/80 \times 100\% = 0.15 \times 100\% = 15\%$.

WHY EACH OPTION IS WRONG:

- A)** This uses the NEW price (68) as the denominator instead of the original price (80), which is the wrong base for a percent decrease calculation.
- B)** This computes the ratio of new to original ($68/80=85\%$) directly, confusing that ratio with the percent decrease itself instead of finding the percent CHANGE.
- C)** This reports only the raw dollar decrease, not the percent the question actually asks for.
- D)** Correct. $(80-68)/80 \times 100\% = 12/80 \times 100\% = 15\%$.

TRAP ALERT

Using the new (smaller) value as the base for the calculation instead of the original (larger, starting) value — percent decrease is always measured relative to where the value STARTED, not where it ended up.

KEY INSIGHT

Percent decrease is always $(\text{original} - \text{new})/\text{original} \times 100\%$ — the ORIGINAL value is always the denominator, never the new value.

FAST METHOD

Find the dollar decrease first ($80 - 68 = 12$), then divide by the STARTING value ($12/80 = 0.15 = 15\%$).

Q24. Correct Answer: **A** • MEDIUM

STEP-BY-STEP SOLUTION:

Translate "a is 150% of b" into an equation: $a = 1.50 \times b$.

Substitute the given value: $90 = 1.50b$.

Divide both sides by 1.50: $b = 90/1.50 = 60$.

WHY EACH OPTION IS WRONG:

A) Correct. $90 = 1.50b$ leads to $b = 90/1.50 = 60$.

B) This multiplies instead of dividing ($90 \times 1.50 = 135$), going in the wrong direction to isolate b.

C) This results from a different incorrect operation (such as multiplying by 0.50 instead of dividing by 1.50).

D) This simply repeats the given value of a instead of solving the equation for b.

TRAP ALERT 🚧

Multiplying by 1.50 instead of dividing by it when solving for b — since a is described as a PERCENT of b (a is the larger, derived quantity), isolating the original quantity b requires division.

KEY INSIGHT 💡

"a is X% of b" always translates to $a = (X/100) \times b$; to solve for b given a, divide a by the decimal form of X%.

FAST METHOD ⚡

Recognize that 150% means a is 1.5 times b, so b is a divided by 1.5: $90/1.5 = 60$.

Q25. Correct Answer: **B** • MEDIUM

STEP-BY-STEP SOLUTION:

Find the tax amount: 8% of \$45 = $0.08 \times 45 = \$3.60$.

Find the tip amount, based on the PRE-TAX amount as specified: 20% of \$45 = $0.20 \times 45 = \$9.00$.

Total = original meal + tax + tip = $45 + 3.60 + 9.00 = \$57.60$.

WHY EACH OPTION IS WRONG:

A) This calculates the tip on the POST-TAX total ($\$48.60 \times 0.20 = \9.72) instead of on the pre-tax amount as the question specifies, leading to a slightly higher total.

B) Correct. $45 + (0.08 \times 45) + (0.20 \times 45) = 45 + 3.60 + 9.00 = 57.60$.

C) This includes the tax but forgets to add the tip entirely.

D) This includes the tip but forgets to add the tax entirely.

TRAP ALERT 🚧

Calculating the tip based on the POST-TAX total instead of the pre-tax amount the question specifically asks for, or forgetting to add one of the two additional charges (tax or tip) entirely.

KEY INSIGHT 💡

When a problem specifies which amount a percentage should be calculated FROM (here, explicitly the pre-tax amount for the tip), that instruction must be followed exactly — tax and tip don't have to be calculated from the same base.

FAST METHOD ⚡

Calculate both the tax and tip separately from the original \$45 (since both are specified or implied to use that base here), then add all three amounts together: original + tax + tip.

Q26. Correct Answer: **360** • MEDIUM  GRID-IN

STEP-BY-STEP SOLUTION:

After the first drain, $100\% - 25\% = 75\%$ of the water remains: $800 \times 0.75 = 600$ liters.

After the second drain, $100\% - 40\% = 60\%$ of the REMAINING 600 liters stays: $600 \times 0.60 = 360$ liters.

COMMON WRONG ANSWERS & WHY:

- Gridding 280 — incorrectly combining the two percentages by simple addition ($25\% + 40\% = 65\%$ drained, so 35% remains: $800 \times 0.35 = 280$) instead of applying them sequentially.
- Gridding 600 — stopping after only the first drain and forgetting to apply the second percentage to the remaining amount.

TRAP ALERT

Adding the two percentages together ($25\% + 40\% = 65\%$ total drained) instead of applying each one sequentially to the progressively smaller remaining amount — the second 40% applies only to what's LEFT after the first drain, not to the original 800 liters.

KEY INSIGHT

When a problem describes successive percent reductions applied to a "remaining" amount, each percentage must be applied to the result of the PREVIOUS step, not to the original starting value or combined through simple addition.

FAST METHOD

Multiply the remaining-fraction factors together in sequence: $800 \times 0.75 \times 0.60 = 360$.

Q27. Correct Answer: **C** • MEDIUM

STEP-BY-STEP SOLUTION:

Let the original price be \$100 for simplicity. After a 25% increase: $100 \times 1.25 = \$125$.

To return to \$100 from \$125, find the required decrease: the decrease amount is $125 - 100 = 25$.

Percent decrease needed = $25/125 \times 100\% = 0.20 \times 100\% = 20\%$ (note: the denominator is 125, the NEW price, since this decrease starts FROM that new price).

WHY EACH OPTION IS WRONG:

- A)** This assumes the percent needed to reverse an increase always equals the original percent increase — but this is only true when the percent change is measured from the SAME base both times, which it isn't here.
- B)** This results from an unrelated incorrect calculation.
- C)** Correct. The decrease needed is $25/125 \times 100\% = 20\%$, since the base for this decrease is the new, larger price of 125, not the original 100.
- D)** This results from a different incorrect base or rounding assumption.

TRAP ALERT

Assuming that "undoing" a percent increase requires the exact same percent decrease — this is a famous trap: since percent increase uses the smaller original value as its base, while the reversing

percent decrease must use the larger new value as ITS base, the two percents are never equal (except for 0%).

KEY INSIGHT

A percent increase and the percent decrease needed to exactly reverse it are NEVER the same number (for any nonzero change) — the required reversing percent is always SMALLER than the original increase, because it's calculated from a larger starting point.

FAST METHOD

After any X% increase, the percent decrease needed to return to the original is $X/(100+X) \times 100\%$: here, $25/125 \times 100\% = 20\%$ — always smaller than X itself.

Q28. Correct Answer: **D** • MEDIUM

STEP-BY-STEP SOLUTION:

Find the growth amount: $8\% \text{ of } 12,000 = 0.08 \times 12,000 = 960$.

Add this growth to the original population: $12,000 + 960 = 12,960$.

WHY EACH OPTION IS WRONG:

- A)** This reports only the AMOUNT of growth (960), not the actual 2021 population the question asks for.
- B)** This results from an arithmetic slip while adding the growth to the original population.
- C)** This results from a different arithmetic slip.
- D)** Correct. $12,000 + (0.08 \times 12,000) = 12,000 + 960 = 12,960$.

TRAP ALERT

Reporting only the growth amount instead of the actual new population, which requires adding that growth back to the original value.

KEY INSIGHT

Population growth (or any percent increase) problems always require two steps: compute the change amount, then add it to the starting value to find the new total.

FAST METHOD

Multiply directly by $(1 + \text{growth rate as a decimal})$: $12,000 \times 1.08 = 12,960$.

Q29. Correct Answer: **A** • HARD

STEP-BY-STEP SOLUTION:

Since the growth compounds each year, multiply by the growth factor (1.10) three separate times — once for each year.

After year 1: $5,000 \times 1.10 = 5,500$. After year 2: $5,500 \times 1.10 = 6,050$. After year 3: $6,050 \times 1.10 = 6,655$.

Equivalently, compute $5,000 \times (1.10)^3 = 5,000 \times 1.331 = 6,655$.

WHY EACH OPTION IS WRONG:

- A)** Correct. $5,000 \times (1.10)^3 = 6,655$.
- B)** This incorrectly treats 3 years of 10% growth as a single flat 30% increase ($10\% \times 3$), instead of compounding the growth year over year.
- C)** This stops after only 2 years of compounding instead of the full 3 years.

D) This accounts for only 1 year of growth instead of all 3 years.

TRAP ALERT 🚫

Treating multi-year percent growth as simple (additive) growth — multiplying the annual percent by the number of years — instead of COMPOUND growth, where each year's growth is calculated on the previous year's already-grown total.

KEY INSIGHT 💡

Repeated percent growth over multiple periods always compounds: multiply by the growth factor once for EACH period, which is equivalent to raising that factor to the power of the number of periods.

FAST METHOD ⚡

Raise the growth factor to the power of the number of years in one step: $5,000 \times (1.10)^3 = 5,000 \times 1.331 = 6,655$.

Q30. Correct Answer: **B** • **HARD**

STEP-BY-STEP SOLUTION:

Let the original price be p . After the 10% increase: $p \times 1.10$. After the subsequent 15% decrease applied to THAT result: $p \times 1.10 \times 0.85$.

Set this equal to the given final price: $p \times 1.10 \times 0.85 = 93.50$.

Simplify the combined factor: $1.10 \times 0.85 = 0.935$.

Solve: $p = 93.50/0.935 = 100$.

WHY EACH OPTION IS WRONG:

A) This results from an arithmetic slip while solving the final division step.

B) Correct. $p \times 0.935 = 93.50$ leads to $p = 93.50/0.935 = 100$.

C) This divides by only one of the two factors (such as just 0.85) instead of the full combined factor (0.935), accounting for only one of the two percent changes.

D) This incorrectly treats the net change as a single flat 5% decrease (15%–10%, found by simple subtraction) instead of properly combining the two changes multiplicatively ($1.10 \times 0.85 = 0.935$, a 6.5% net decrease, not 5%).

TRAP ALERT 🚫

Combining two successive percent changes by simple subtraction (15%–10%=5% net decrease) instead of multiplying the two growth/decay factors together — these give different results because percent changes compound multiplicatively, not additively.

KEY INSIGHT 💡

To reverse two successive percent changes, multiply the two factors together first to get a single combined factor, then divide the final value by that ONE combined factor — never combine the original percents by addition or subtraction.

FAST METHOD ⚡

Combine the two factors first ($1.10 \times 0.85 = 0.935$), then divide the final value by that single combined number: $93.50 \div 0.935 = 100$.

Q31. Correct Answer: **50** • **HARD** 🖊️ **GRID-IN**

STEP-BY-STEP SOLUTION:

Translate each relationship into an equation: $x = 1.20y$ (since x is 20% greater than y), and $y = 1.25z$ (since y is 25% greater than z).

Substitute the second equation into the first: $x = 1.20 \times (1.25z) = 1.50z$.

Since $x = 1.50z$, x is 50% greater than z (because $1.50 = 1 + 0.50$, representing a 50% increase over z).

COMMON WRONG ANSWERS & WHY:

- Gridding 45 — incorrectly adding the two percents directly ($20\% + 25\% = 45\%$) instead of properly combining the two multiplicative relationships.
- Gridding 1.5 — reporting the combined growth FACTOR (1.50) itself instead of converting it to the percent increase the question actually asks for (50%).

TRAP ALERT

Adding the two given percentages directly ($20\% + 25\% = 45\%$) instead of recognizing that two successive "X% greater than" relationships must be COMBINED MULTIPLICATIVELY (by multiplying their growth factors), not added.

KEY INSIGHT

When quantities are related by chained percent relationships (x in terms of y , y in terms of z), substitute one equation into the other and multiply the growth FACTORS together — the combined percent increase is always slightly more than the simple sum of the individual percents.

FAST METHOD

Convert each relationship to a decimal growth factor (20% greater $\rightarrow 1.20$, 25% greater $\rightarrow 1.25$), multiply the factors together ($1.20 \times 1.25 = 1.50$), then convert back to a percent increase (1.50 means 50% greater).

Q32. Correct Answer: C • HARD

STEP-BY-STEP SOLUTION:

Percent error is calculated as $|\text{measured value} - \text{actual (intended) value}| / (\text{actual value}) \times 100\%$.

$|263 - 250| = 13$.

Percent error = $13/250 \times 100\% = 0.052 \times 100\% = 5.2\%$.

WHY EACH OPTION IS WRONG:

- A) This results from an arithmetic slip while dividing.
- B) This rounds to a nearby "clean-looking" number instead of computing the precise value, losing accuracy in the process.
- C) Correct. $|263 - 250| / 250 \times 100\% = 13/250 \times 100\% = 5.2\%$.
- D) This reports the raw gram difference (13) as if it were already the percent error, without ever dividing by the intended (actual) value.

TRAP ALERT

With numbers that don't form an obviously "clean" percent, it's tempting to round the raw difference into a nearby familiar-looking percent instead of carrying out the full, precise division.

KEY INSIGHT

Percent error calculations don't always produce round numbers — trust the full calculation (difference $\div$ actual value $\times 100\%$) rather than estimating or rounding prematurely, especially when the question asks for a specific decimal precision.

FAST METHOD

Compute the difference first ($263 - 250 = 13$), then divide carefully by the actual/intended value ($13/250 = 0.052 = 5.2\%$) using a calculator for precision.

Q33. Correct Answer: **D • HARD**

STEP-BY-STEP SOLUTION:

Let the unknown number be n . Translate the relationship into an equation: $0.30n = 0.50n - 18$ (since 30% of n is 18 LESS than 50% of n).

Rearrange to isolate n : $18 = 0.50n - 0.30n$.

$18 = 0.20n$.

$n = 18/0.20 = 90$.

WHY EACH OPTION IS WRONG:

A) This incorrectly sets up the equation as $0.30n = 18$ directly, ignoring that the relationship is actually a DIFFERENCE between 50% and 30% of the number, not just 30% alone equaling 18.

B) This results from an unrelated incorrect setup, such as dividing 18 by an incorrect percent.

C) This results from a different arithmetic slip in setting up or solving the equation.

D) Correct. $0.50n - 0.30n = 18$ leads to $0.20n = 18$, so $n = 90$.

TRAP ALERT 🚧

Misreading the relationship and setting up the equation as if 30% of the number equals 18 directly, instead of correctly recognizing that the DIFFERENCE between 50% and 30% of the number equals 18.

KEY INSIGHT 💡

When a problem describes one percentage of a number as being a certain amount MORE or LESS than another percentage of that SAME number, the equation must capture the DIFFERENCE between the two percentages, not treat either percentage as standing alone.

FAST METHOD ⚡

Recognize that the difference between the two given percentages ($50\% - 30\% = 20\%$) directly corresponds to the given amount (18), so $0.20n = 18$, giving $n = 18/0.20 = 90$ in one clean step.

Q34. Correct Answer: **A • HARD**

STEP-BY-STEP SOLUTION:

Since depreciation compounds each year, multiply by the decay factor (0.85, since $100\% - 15\% = 85\%$) twice — once for each year.

After year 1: $24,000 \times 0.85 = 20,400$. After year 2: $20,400 \times 0.85 = 17,340$.

Equivalently, compute $24,000 \times (0.85)^2 = 24,000 \times 0.7225 = 17,340$.

WHY EACH OPTION IS WRONG:

A) Correct. $24,000 \times (0.85)^2 = 17,340$.

B) This incorrectly treats 2 years of 15% depreciation as a single flat 30% decrease ($15\% \times 2$), instead of compounding the depreciation year over year.

C) This accounts for only 1 year of depreciation instead of both years.

D) This results from a different arithmetic slip in the compounding calculation.

TRAP ALERT 🚧

Treating multi-year percent depreciation as simple (additive) decay — multiplying the annual percent by the number of years — instead of COMPOUND decay, where each year's depreciation is calculated on the previous year's already-reduced value.

KEY INSIGHT 💡

Repeated percent depreciation, just like repeated growth, always compounds: multiply by the decay factor once for EACH period, which is equivalent to raising that factor to the power of the number of periods.

FAST METHOD ⚡

Raise the decay factor to the power of the number of years in one step: $24,000 \times (0.85)^2 = 24,000 \times 0.7225 = 17,340$.

Q35. Correct Answer: 10 • HARD 🖊️ GRID-IN

STEP-BY-STEP SOLUTION:

Percent decrease is calculated as $(\text{original} - \text{new})/(\text{original}) \times 100\%$, where the relevant "original" for THIS particular change is \$150 (the starting point of the decrease being asked about).

The decrease amount is $150 - 135 = 15$.

Percent decrease = $15/150 \times 100\% = 0.10 \times 100\% = 10\%$.

COMMON WRONG ANSWERS & WHY:

- Gridding 11.1 — using the final value (135) as the denominator instead of the value the decrease started FROM (150).
- Gridding 15 — reporting the raw dollar decrease instead of converting it to a percent of the correct base value.

TRAP ALERT 🚨

With multiple dollar values mentioned in the problem (an original price, \$150, and \$135), it's easy to use the wrong pair or the wrong base when the question specifically asks about only ONE particular change (from \$150 to \$135, not from the very original price).

KEY INSIGHT 💡

When a problem describes multiple sequential changes, carefully identify exactly which TWO values the specific question is asking about, and use the EARLIER of those two values as the base for that particular percent calculation.

FAST METHOD ⚡

Identify the two relevant values for the specific change asked about (150 to 135), then compute $(\text{difference})/(\text{starting value}) \times 100\%$ using only those two numbers: $15/150 = 10\%$.

Q36. Correct Answer: B • HARD

STEP-BY-STEP SOLUTION:

First find the number of students who prefer math: $60\% \text{ of } 400 = 0.60 \times 400 = 240$ students.

Then find 25% of THIS group (the math-preferring students), since the second percentage applies only to that subgroup, not the full 400: $0.25 \times 240 = 60$ students.

WHY EACH OPTION IS WRONG:

A) This incorrectly applies the second percentage (25%) directly to the ORIGINAL 400 students, instead of to the smaller math-preferring subgroup of 240.

- B)** Correct. $0.60 \times 400 = 240$ (math), then $0.25 \times 240 = 60$ (both math and science).
C) This stops after finding only the math-preferring count (240), forgetting to apply the second percentage at all.
D) This results from a different arithmetic slip in combining the two percentages.

TRAP ALERT 🚧

Applying the second percentage (25%) to the original total (400 students) instead of to the SUBGROUP it actually describes (the 240 students who prefer math) — the phrase "of the students who prefer math" signals a smaller, specific base.

KEY INSIGHT 💡

When a percentage is described as being "of" a specific subgroup (not the original total), that subgroup — not the original total — becomes the new base for that calculation.

FAST METHOD ⚡

Work through the percentages in the exact order they're described, using each new result as the base for the NEXT percentage: $400 \rightarrow (\times 0.60) \rightarrow 240 \rightarrow (\times 0.25) \rightarrow 60$.

Q37. Correct Answer: **C** • **HARD**

STEP-BY-STEP SOLUTION:

Find the combined remaining-value factor by multiplying the two individual remaining-value factors: $(1 - 0.20) \times (1 - 0.10) = 0.80 \times 0.90 = 0.72$.

This means 72% of the original price remains after both discounts, so the equivalent SINGLE discount is $100\% - 72\% = 28\%$.

WHY EACH OPTION IS WRONG:

- A)** This simply adds the two discount percentages together ($20\% + 10\% = 30\%$), which is NOT how successive discounts combine — the second discount is applied to an already-reduced price, making the true combined discount slightly less than the simple sum.
B) This results from a different incorrect combination of the two percentages.
C) Correct. $(0.80)(0.90) = 0.72$ remains, so the equivalent single discount is $100\% - 72\% = 28\%$.
D) This multiplies the two discount RATES directly ($0.20 \times 0.10 = 0.02$), instead of multiplying the REMAINING-value factors (0.80×0.90) and then converting back to a discount percent.

TRAP ALERT 🚧

Adding two successive discount percentages together as if they combine additively ($20\% + 10\% = 30\%$) — this is ALWAYS an overestimate, since the second discount applies to a smaller, already-discounted price, making the true combined discount slightly less than the simple sum.

KEY INSIGHT 💡

Successive discounts never simply add together. To find the single equivalent discount, multiply the REMAINING-value factors (1 minus each discount) together, then subtract that product from 100% to get the true combined discount.

FAST METHOD ⚡

Multiply the two "remaining" factors directly: $0.80 \times 0.90 = 0.72$, meaning 72% remains, so the discount is $100\% - 72\% = 28\%$ — always a bit less than simply adding the two original percents.

Q38. Correct Answer: **D** • **HARD**

STEP-BY-STEP SOLUTION:

Translate the relationship into an equation: $0.60x = 0.75y$.

Substitute the given value of y : $0.60x = 0.75(80) = 60$.

Divide both sides by 0.60: $x = 60/0.60 = 100$.

WHY EACH OPTION IS WRONG:

- A)** This correctly computes the right-hand side of the equation ($0.75 \times 80 = 60$) but then stops there, forgetting to actually divide by 0.60 to solve for x .
- B)** This results from an unrelated incorrect operation, such as multiplying y by an incorrect factor.
- C)** This results from a different unrelated incorrect operation.
- D)** Correct. $0.60x = 0.75(80) = 60$ leads to $x = 60/0.60 = 100$.

TRAP ALERT

Correctly computing the value of 75% of y , but then mistakenly treating THAT intermediate result as the final answer for x , instead of using it to complete the final division step required to actually solve for x .

KEY INSIGHT

When a problem states that one percentage of an unknown equals another percentage of a known value, first compute the known side completely, then solve the resulting one-step equation for the unknown — don't stop at the intermediate value.

FAST METHOD

Compute the known side fully first ($0.75 \times 80 = 60$), then immediately divide by the OTHER percentage's decimal form to isolate x : $60/0.60 = 100$.

Q39. Correct Answer: **100000** • **HARD**  **GRID-IN**

STEP-BY-STEP SOLUTION:

Let the original revenue be r . After the 15% increase in Year 1: $r \times 1.15$. After the 10% decrease in Year 2, applied to THAT result: $r \times 1.15 \times 0.90$.

Set this equal to the given Year 2 revenue: $r \times 1.15 \times 0.90 = 103,500$.

Simplify the combined factor: $1.15 \times 0.90 = 1.035$.

Solve: $r = 103,500/1.035 = 100,000$.

COMMON WRONG ANSWERS & WHY:

- Gridding 105679 (approximately) — incorrectly combining the two percent changes by simple subtraction ($15\% - 10\% = 5\%$ net increase) and dividing by 1.05 instead of by the correct combined factor 1.035.
- Gridding 90000 — dividing by only one of the two factors (such as just 1.15) instead of the full combined factor.

TRAP ALERT

Combining the 15% increase and 10% decrease by simple subtraction (treating it as a flat 5% net increase) instead of properly multiplying the two factors together — the true combined factor (1.035, a 3.5% net increase) is different from the naive 5% estimate.

KEY INSIGHT

Two different successive percent changes (one increase, one decrease) must be combined by MULTIPLYING their respective factors together, exactly like any other successive percent change problem — never by subtracting the raw percents from each other.

FAST METHOD ⚡

Combine the two factors first ($1.15 \times 0.90 = 1.035$), then divide the final value by that single combined number: $103,500 \div 1.035 = 100,000$.

Q40. Correct Answer: **A** • **HARD**

STEP-BY-STEP SOLUTION:

Let the Year 1 profit be p . After the 40% increase: $p \times 1.40$. After the subsequent 30% decrease applied to THAT result: $p \times 1.40 \times 0.70$.

Set this equal to the given Year 3 profit: $p \times 1.40 \times 0.70 = 58,800$.

Simplify the combined factor: $1.40 \times 0.70 = 0.98$.

Solve: $p = 58,800 / 0.98 = 60,000$.

WHY EACH OPTION IS WRONG:

A) Correct. $p \times 0.98 = 58,800$ leads to $p = 58,800 / 0.98 = 60,000$.

B) This incorrectly multiplies the Year 3 profit by the combined factor again ($58,800 \times 0.98$) instead of dividing by it, moving in the wrong direction entirely.

C) This divides by only one of the two factors (0.70) instead of the full combined factor (0.98), accounting for only the decrease and ignoring the increase.

D) This multiplies by only one of the two factors (0.70) instead of properly dividing by the full combined factor.

TRAP ALERT 🚨

With a combined factor (0.98) that's very close to 1, it's tempting to assume the two percent changes nearly cancel out and the answer should be very close to \$58,800 itself — but the precise calculation must still be carried out completely.

KEY INSIGHT 💡

Even when a 40% increase and a 30% decrease seem like they might roughly offset each other, the actual combined factor ($1.40 \times 0.70 = 0.98$, just a 2% net decrease) must be calculated precisely — don't substitute intuition for the actual multiplication.

FAST METHOD ⚡

Combine the two factors first ($1.40 \times 0.70 = 0.98$), then divide the final value by that single combined number: $58,800 \div 0.98 = 60,000$.

TOPIC SUMMARY — Percentages

Common Mistakes:

1. Adding successive percent changes instead of multiplying their factors: a 20% increase followed by a 20% decrease is NEVER a net 0% change — it's always a small net decrease. Two percent changes always compound multiplicatively, so the correct approach is $(1 + \text{first change}) \times (1 + \text{second change})$, never simply adding the raw percentages.
2. Reversing a percent change by applying that same percentage in the opposite direction: to undo a 15% increase, dividing the new value by 1.15 is correct — but simply taking 15% OFF the new value is NOT equivalent, since the second percentage would be calculated from a different (larger) base.
3. Using the wrong base value when calculating a percent change, error, or markup: the base is ALWAYS the original, starting, or reference value — the one being compared FROM, or the one

right after the word 'than' in the problem. Using the final value, a sum, or an incorrect reference value as the denominator is one of the most common sources of wrong answers on percent questions.

Must Remember:

1. Percent of a number: convert to decimal first, then multiply. To find the original given a percent result: divide the final value by the decimal form of the percent (e.g., if 115% of the original = 138, then original = $138 \div 1.15 = 120$).
2. Successive percent changes compound multiplicatively: combine by multiplying the decimal factors together. Two successive discounts of 20% and 10% leave $0.80 \times 0.90 = 72\%$ of the price, equivalent to a single 28% discount — never 30%.
3. Percent increase/decrease formula: $(\text{change} \div \text{original}) \times 100\%$, where 'original' is always the STARTING value for that specific change. Percent error uses the same formula: $(|\text{measured} - \text{actual}| \div \text{actual}) \times 100\%$.

Expected on DSAT:

Based on this project's frequency analysis of past Digital SAT papers, percentages appeared in 11 of 1,240 math questions analyzed (about 0.9%). That works out to roughly 0–1 question per test on average. When the topic does appear, it tends to involve at least one non-trivial twist — either a two-step percent change (increase then decrease, or reverse-find-the-original), a percent-of-a-percent, or a successive-discount trap — rather than a straightforward single-step calculation.

— END OF SOLUTIONS —